Why Warmup Helps: A Non-Asymptotic Analysis of the WSD Schedule under (H_0, H_1)-smoothness and Aiming Condition

WONG, Hok Fong Silas

Machine Learning and Optimization Laboratory, EPFL

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Checklist (as of Aug 25, 2026)

- Warmup v.s. constant SGD: statement (easy/medium)
- Remark: recovering GD with $\sigma_\delta \to 0, H_1 \to 0$ (medium)
- Sample-loss-based surrogate: statement + proof (medium, note that this can approximately recover the empirical schedule of Alimisis et al., 2025)
- Noise models: statement + proof (hard)
- Discussions: (1) Different cooldown shapes (1-sqrt, linear, cosine); (2) Stable is needed when...?; (3) Optimizing the bounds (η_{peak}, lowered-linear $c_0^\star$, PL-type bound last iterate). (easy; easy; medium); (4) Recent work: WSqD
- Numerical experiments: design + statements (hard)
- Approximation: Does the saturating warmup schedule resemble the exponential warmup? Does the inverse-sqrt-decay schedule mimic the linear cooldown to some extent? (medium)
- Rebuttal: understanding the connection with and dissimilarity to EoS theory; extension to μ-PL structures (likely easy just tedious)

Highlights

Focus: optimization viewpoint for the special class of functions. Justifications of the assumptions are mostly compensated by previous works.

- Rigorous linear warmup analysis and fair comparison with constant stepsize
- Clear development of theory of SGD inspired by noiseless GD settings
- Relaxation of assumptions, e.g. interpolation condition, convexity (caveat: we have assumed properties of the loss landscape that can be viewed as a case of convexity), full knowledge about the value of $f^\star$ and f_{S_k}
- One reason for lowering the peak learning rate under longer training horizon: bounding noise amplification to keep iterates around the basin
- Plug-in-and-play decaying stepsizes
- Self-tuning warmup strategy theoretically (much) better (Experiments: To do!)
- Quantification of warmup shapes and stage partitioning, connecting theory with practice

Contents

Setup and Notations

Assumptions

Assumption 1 (Generalized Smoothness)

The function $f : \mathbb{R}^d \to \mathbb{R}$ *with* $f^\star := \inf_{w \in \mathbb{R}^d} f(\mathbf{w}) > -\infty$ *is twice differentiable and* (ρ, H_0, H_ρ)*-smooth, i.e.*

$$\|\nabla^2 f(\mathbf{w})\| \leq H_0 + H_\rho (f(\mathbf{w}) - f^\star)^\rho, \qquad H_0, H_\rho \geq 0, \rho > 0.$$

Throughout we take $\rho = 1$ *and write* $H_1 := H_\rho$*.*

Assumption 2 (Aiming Condition)

There is $\theta > 0$ *such that, with* $\mathcal{S} := \arg\min_{\mathbf{w} \in \mathbb{R}^d} f(\mathbf{w})$*,*

$$\langle \nabla f(\mathbf{w}), \mathbf{w} - \pi_{\mathcal{S}}(\mathbf{w}) \rangle \geq \theta (f(\mathbf{w}) - f^\star) \qquad \forall \mathbf{w} \in \mathbb{R}^d.$$

Assumption 3 (Sub-Gaussian Noise)

The stochastic noise $\boldsymbol{\xi}_k$ *has* $\mathbb{E}[\boldsymbol{\xi}_k \mid \mathcal{F}_k] = 0$ *and*

$$\mathbb{P}(\|\boldsymbol{\xi}_k\| \geq t \mid \mathcal{F}_k) \leq 2\exp(-t^2/(2\sigma^2)), \qquad t \geq 0,$$

where $\mathcal{F}_k := \sigma(\mathbf{w}_0, \ldots, \mathbf{w}_k)$.

Definition 1 (WSD Schedule)

Given $T := T_w + T_s + T_d$ *(warmup, stable, decay), a monotonically decreasing* $\psi : [0,1] \to [0,1]$ *with* $\psi(0) = 1$, $\psi(1) = 0$, *and* $0 < \eta_{\text{start}} \leq \eta_{\text{peak}}$, *set*

$$\eta_k := \begin{cases} \eta_{\text{start}} + \dfrac{k}{T_w}(\eta_{\text{peak}} - \eta_{\text{start}}), & 0 \leq k < T_w, \\ \eta_{\text{peak}}, & T_w \leq k < T_w + T_s, \\ \eta_{\text{peak}}\psi\big(\frac{k - T_w - T_s}{T_d}\big), & T_w + T_s \leq k \leq T. \end{cases}$$

Moreover, we write $S_k := \sum_{j<k} \eta_j$.

Warmup Provably Helps: Analysis of GD

Analysis of GD

We consider the deterministic gradient descent algorithm, where

$$\mathbf{w}_{k+1} = \mathbf{w}_k - \eta_k \nabla f(\mathbf{w}_k), \Delta_k := f(\mathbf{w}_k) - f^\star, D_k := \mathsf{dist}(\mathbf{w}_k, \mathcal{S}) = \|\mathbf{w}_k - \pi_{\mathcal{S}}(\mathbf{w}_k)\|.$$

Theorem 1 (GD Post-Warmup Last-Iterate Guarantees)

Suppose that Assumptions 1–2 holds. Let $U \geq \Delta_0$ be any **upper bound** *on the initial suboptimality gap, $c \in (0, 1/3]$ be a constant and $\{\mathbf{w}_k\}$ be generated by GD, with the learning-rate schedule of Definition 1, where*

$$\eta_{\text{start}} := \frac{c}{2(H_0 + H_1 U)}, \quad \eta_{\text{peak}} := \frac{c}{2H_0}, \quad \widetilde{D} := \sqrt{D_0^2 + \frac{cU}{2(1-c)H_0}},$$

and

$$T_w \geq \left\lceil 1 + \frac{H_1 U}{H_0} \right\rceil \cdot \max\left\{ \left\lceil \frac{36 H_1 \widetilde{D}^2}{c\theta^2} \right\rceil, \left\lceil \frac{24}{c(1-c)} \right\rceil \right\}.$$

Analysis of GD

Theorem 1 (GD Post-Warmup Last-Iterate Guarantees)

With $\alpha := (1-c)\cdot\min\{\theta^2/\widetilde{D}^2, H_1\}$, $\Delta_{T_w} \le H_0/(3H_1)$, *and for every* $T_w < k \le T_w + T_s$, *we have*

$$\Delta_k \le \frac{2H_0}{\alpha c(k-T_w)}, \quad \|\nabla f(\mathbf{w}_k)\|^2 \le \frac{9H_0^2}{2\alpha c(k-T_w)} + \frac{27H_0^2 H_1}{\alpha^2 c^2 (k-T_w)^2};$$

In contrast, for the iterates $\{\mathbf{w}_k\}$ *generated by constant-stepsize GD with the largest admissible step* $\eta_c = \eta_{\text{start}}$, *by setting* $\widetilde{D}_c := \sqrt{D_0^2 + \eta_c U/(1-c)} \le \widetilde{D}$ *and* $\alpha_c := (1-c)\min\{\theta^2/\widetilde{D}_c^2, H_1\} \ge \alpha$, *we have the guarantee that*

$$\Delta_k \le \frac{2(H_0+H_1U)}{\alpha_c ck}, \quad \|\nabla f(\mathbf{w}_k)\|^2 \le \frac{9H_0(H_0+H_1U)}{2\alpha_c ck} + \frac{27H_1(H_0+H_1U)^2}{\alpha_c^2 c^2 k^2},$$

for all $k \ge 1$.

Spending Negligible Iterations on Warmup Is Indeed Better

Remark 1 (Last-Iterate Comparison)

From Theorem 1, we can conduct a comparison of the last-iterate bounds. For $k \gg T_w$, the guarantees provided by the two schedules differ only in H_0 versus $H_0 + H_1\Delta_0$ and in α versus α_c. We consider the suboptimality gap bounds and the leading $1/k$ term of the squared gradient bounds, where we have

$$\frac{\Delta_k^{\textit{const}}}{\Delta_k^{\textit{warmup}}}, \frac{\|\nabla f(\mathbf{w}_k)\|^2_{\textit{const}}}{\|\nabla f(\mathbf{w}_k)\|^2_{\textit{warmup}}} \to \frac{(H_0 + H_1\Delta_0)\alpha}{H_0\alpha_c} = \left(1 + \frac{H_1\Delta_0}{H_0}\right)\frac{\alpha}{\alpha_c} \geq 1.$$

Warmup schedule is always no worse than the constant-stepsize schedule, with respect to the ratio above.

Uniform Localization under Mean-Zero Noise: Analysis of SGD

Analysis of SGD

Assumption 4 (Stage Lengths)

Let $U \geq \Delta_0$ be any **upper bound** *on the initial suboptimality gap. Suppose Assumptions 1–3 holds. Let $c \in (0, 1/36]$ be a constant, $\delta \in (0,1)$ be the failure probability, $\Delta_0 := f(\mathbf{w}_0) - f^\star$, and $\sigma_\delta := \sigma\sqrt{2\ln(4T/\delta)}$. Using the WSD schedule as in Definition 1 and fix the constants*

$$\eta_{\text{start}} := \min\left\{\frac{c}{2(H_0 + H_1 U)}, \frac{c}{2M}\right\}, \quad \eta_{\text{peak}} := \min\left\{\frac{c}{2H_0}, \frac{c}{2M}\right\},$$

$$\alpha := \frac{1-2c}{2}\min\left\{\frac{\theta^2}{4\widetilde{D}^2}, H_1\right\},$$

$$M := 4\sigma_\delta H_1 \max\left\{\frac{2\widetilde{D}}{\theta}, \frac{1}{\sqrt{H_1}}\right\}, \quad \widetilde{D} := \sqrt{D_0^2 + \frac{U}{4H_0}},$$

$$\Delta_{\text{th}} := \sqrt{\frac{1+4c(1-2c)}{(1-2c)^2}} \cdot \sigma_\delta \cdot \max\left\{\frac{2\widetilde{D}}{\theta}, \frac{1}{\sqrt{H_1}}\right\}.$$

Analysis of SGD

Assumption 4 (Stage Lengths)

We require the stage lengths to satisfy

$$T_w \geq \max\left\{2, \min\left\{\left\lceil \frac{64H_1}{\alpha c} \right\rceil, \left\lceil \frac{H_1 \widetilde{D}^2}{\theta^2} \right\rceil\right\}\right\} \cdot \left\lceil \frac{H_0 + H_1 U}{M} \right\rceil.$$

Theorem 2 (Convergence Properties of the WSD Schedule with Mean-Zero Noise)

Fix $c \in (0, 1/36]$, $\delta \in (0,1)$, *and suppose that Assumptions 1–3 holds. Let* $\{\mathbf{w}_k\}$ *be generated by SGD, with the learning-rate schedule of Definition 1 satisfying Assumption 4 and*

$$V_T := \sum_{k<T} \eta_k^2 \leq \frac{\widetilde{D}^2}{1536\sigma_\delta^2 \ln(2/\delta)}.$$

Analysis of SGD

Theorem 2 (Convergence Properties of the WSD Schedule with Mean-Zero Noise)

Then, there is an event $\mathcal{E}$ with $\mathbb{P}(\mathcal{E}) \geq 1 - \delta$ such that $\Delta_{T_w} \leq \max\{2\Delta_{\mathrm{th}}, H_0/H_1\}$, the minimum gradient satisfies

$$\mathbb{E}\left[\min_{T_w \leq k \leq T} \|\nabla f(\mathbf{w}_k)\|^2 \middle| \mathcal{E}\right] \leq \frac{1}{(1-c)\mathbb{P}(\mathcal{E})}\left(\frac{\mathbb{E}[\Delta_{T_w}]}{\sum_{k=T_w}^{T-1} \eta_k} + (H_0 + 2H_1\Delta_{\mathrm{th}})\sigma_\delta^2 \frac{\sum_{k=T_w}^{T-1} \eta_k^2}{\sum_{k=T_w}^{T-1} \eta_k}\right);$$

Moreover, the function values satisfy

$$\frac{\sum_{k=T_w}^{T-1} \eta_k \Delta_k}{\sum_{k=T_w}^{T-1} \eta_k} \leq \frac{5\widetilde{D}^2 + \frac{8}{1-2c}\eta_{\mathrm{peak}}\Delta_{\mathrm{th}}}{2\theta \sum_{k=T_w}^{T-1} \eta_k}.$$

Analysis of SGD

Theorem 2 (Convergence Properties of the WSD Schedule with Mean-Zero Noise)

Furthermore, using the decay profile $\psi(x) := 1/\sqrt{1 + xT_d}$, so that for k in the decay phase $\eta_k = \eta_{\mathrm{peak}}(1 + (k - T_w - T_s))^{-1/2}$, we have that

$$\mathbb{E}\left[\min_{T_w \le k \le T} \|\nabla f(\mathbf{w}_k)\|^2 \middle| \mathcal{E}\right] \le \frac{1}{(1-c)\mathbb{P}(\mathcal{E})}\left(\frac{2\Delta_{\mathrm{th}}}{\eta_{\mathrm{peak}}(T_s + \sqrt{T_d})} + \frac{(H_0 + 2H_1\Delta_{\mathrm{th}})\sigma_\delta^2 \eta_{\mathrm{peak}}(T_s + 1 + \ln T_d)}{T_s + \sqrt{T_d}}\right),$$

$$\frac{\sum_{k=T_w}^{T-1} \eta_k \Delta_k}{\sum_{k=T_w}^{T-1} \eta_k} \le \frac{1}{2\theta(T_s + \sqrt{T_d})} \cdot \left(\frac{5\widetilde{D}^2}{\eta_{\mathrm{peak}}} + \frac{8\Delta_{\mathrm{th}}}{1-2c}\right).$$

Intended improvement: *we assume that Δ_k is noise-limited, where the best suboptimality gap is eventually constrained by the noise bound, i.e., Δ_{th}. This is not really a concern for small σ and it would cause the warmup length to blowup. To completely reduce to the GD case using $\sigma \to 0$, we also need to incorporate the curvature limited threshold, i.e., H_0/H_1. We are currently updating the proof to accommodate this situation (T_w and Δ_{T_w} bound).*

Iteration Complexities

Remark 2 (Explicit Regimes for Fully-Resolved Iteration Complexities)

Let T_w be the number of iterations for the suboptimality gap to reach the noise floor, T_d^{grad} be the number of iterations to guarantee that $\mathbb{E}[\min_{T_w \le k < T} \|\nabla f(\mathbf{w}_k)\|^2 \mid \mathcal{E}] \le \varepsilon_g^2$ and T_d^{subopt} be the number of iterations to guarantee that $\sum_{k=T_w}^{T-1} \eta_k \Delta_k / \sum_{k=T_w}^{T-1} \eta_k \le \varepsilon_s$.
We resolve the $\min$ and $\max$ operators by considering the two branches of the contraction coefficient α and the two branches of the peak-step cap η_{peak} in the iteration complexity bounds and provide explicit stage-length formulas for four regimes as follows:
Contraction branches. From $\alpha = (1-2c)\min\{\theta^2/(4\widetilde{D}^2), H_1\}$, we consider two conditions: $\theta^2/(4\widetilde{D}^2) \le H_1$ (A1, aiming-limited) and $\theta^2/4\widetilde{D}^2 > H_1$ (A2, smoothness-limited).
Peak-step branches. From $\eta_{\mathrm{peak}} = c/(2\max\{H_0, M\})$, we also have two conditions: $H_0 \ge M$ (E1, smoothness-capped), i.e. $\eta_{\mathrm{peak}} = c/(2H_0)$, and $H_0 < M$ (E2, noise-capped), i.e. $\eta_{\mathrm{peak}} = c/(2M)$.

Iteration Complexities

Remark 2 (Explicit Regimes for Fully-Resolved Iteration Complexities (cont.))

Regime	Conditions	Warmup T_w	Min. Gradient T_d^{grad}	Weighted Subopt. T_d^{subopt}
1	$H_0 \geq \frac{8\sigma_\delta H_1 \widetilde{D}}{\theta}$, $\theta \leq 2\widetilde{D}\sqrt{H_1}$	$\widetilde{\Theta}\left(\frac{(H_0+H_1\Delta_0)\widetilde{D}^2}{\sigma_\delta H_1 \theta^2}\right)$	$\widetilde{\Theta}\left(\frac{H_0^2 \widetilde{D}^2 \sigma_\delta^2}{\theta^2 \varepsilon_g^4 (1-\delta)^2}\right)$	$\widetilde{\Theta}\left(\frac{H_0^2 \widetilde{D}^4}{\theta^2 \varepsilon_s^2}\right)$
2	$H_0 < \frac{8\sigma_\delta H_1 \widetilde{D}}{\theta}$, $\theta \leq 2\widetilde{D}\sqrt{H_1}$	$\widetilde{\Theta}\left(\frac{(H_0+H_1\Delta_0)\widetilde{D}^2}{\sigma_\delta H_1 \theta^2}\right)$	$\widetilde{\Theta}\left(\frac{H_1^2 \widetilde{D}^4 \sigma_\delta^4}{\theta^4 \varepsilon_g^4 (1-\delta)^2}\right)$	$\widetilde{\Theta}\left(\frac{H_1^2 \widetilde{D}^6 \sigma_\delta^2}{\theta^4 \varepsilon_s^2}\right)$
3	$H_0 \geq 4\sigma_\delta\sqrt{H_1}$, $\theta > 2\widetilde{D}\sqrt{H_1}$	$\widetilde{\Theta}\left(\frac{H_0+H_1\Delta_0}{\sigma_\delta\sqrt{H_1}}\right)$	$\widetilde{\Theta}\left(\frac{H_0^2 \sigma_\delta^2}{H_1 \varepsilon_g^4 (1-\delta)^2}\right)$	$\widetilde{\Theta}\left(\frac{H_0^2 \widetilde{D}^4}{\theta^2 \varepsilon_s^2}\right)$
4	$H_0 < 4\sigma_\delta\sqrt{H_1}$, $\theta > 2\widetilde{D}\sqrt{H_1}$	$\widetilde{\Theta}\left(\frac{H_0+H_1\Delta_0}{\sigma_\delta\sqrt{H_1}}\right)$	$\widetilde{\Theta}\left(\frac{\sigma_\delta^4}{\varepsilon_g^4 (1-\delta)^2}\right)$	$\widetilde{\Theta}\left(\frac{H_1 \widetilde{D}^4 \sigma_\delta^2}{\theta^2 \varepsilon_s^2}\right)$

Table: Stage lengths by regime.

Self-Tuning Schedules

Self-Tuning Schedules: GD

Definition 2 (Saturating Warmup for GD)

Suppose that Assumptions 1–2 hold. Let $U \geq \Delta_0$ *be any upper bound on the initial suboptimality gap and let* $c \in (0, 1/3]$, $\alpha = (1-c)\min\{\theta^2/\widetilde{D}_0^2, H_1\}$, *where* $\widetilde{D} := \sqrt{D_0^2 + \frac{cU}{(1-c)H_0}}$. *Set* $\widehat{\Delta}_0 := U$ *and for all* $k \geq 0$, *let*

$$\eta_k := \frac{c}{H_0 + H_1 \widehat{\Delta}_k}, \quad \widehat{\Delta}_{k+1} := \frac{\widehat{\Delta}_k}{1 + \alpha \eta_k \widehat{\Delta}_k}.$$

Theorem 3 (Guarantees of GD under the Self-Tuning Warmup Schedule)

Suppose Assumptions 1–2 hold and fix $c \in (0, 1/3]$. *Let* $\{\mathbf{w}_k\}$ *be generated by GD using the schedule as in Definition 2, the suboptimality gap reaches the post-warmup benign-curvature state* $\widehat{\Delta}_k \leq H_0/H_1$ *after*

$$T_w \geq \max\left\{0, \left\lceil 1 + \ln\left(\frac{H_1 U}{H_0}\right)\right\rceil \cdot \max\left\{\left\lceil \frac{1}{2c(1-c)}\right\rceil, \left\lceil \frac{H_1 \widetilde{D}^2}{2c(1-c)\theta^2}\right\rceil\right\}\right\}$$

steps, which is logarithmic in $H_1 U / H_0$ *compared to Theorem 1.*

Self-Tuning Schedules: SGD

Definition 3 (Saturating Warmup for SGD)

Suppose that Assumptions 1–3 hold. Let $U \geq \Delta_0$ *be any upper bound on the initial suboptimality gap and let* $c \in (0, 1/36]$,

$$\alpha := \frac{1-2c}{2} \min\left\{\frac{\theta^2}{4\widetilde{D}^2}, H_1\right\}, \quad \widetilde{D}_0 := \sqrt{D_0^2 + \frac{U}{4H_0}}.$$

Set $\widehat{\Delta}_0 := U$ *and for all* $k \geq 0$, *let*

$$\eta_k := \frac{c}{H_0 + H_1 \widehat{\Delta}_k}, \quad \widehat{\Delta}_{k+1} := \max\left\{\widehat{\Delta}_k - \alpha\eta_k(\widehat{\Delta}_k^2 - \Delta_{\mathrm{th}}^2), \Delta_{\mathrm{th}}\right\},$$

where

$$\Delta_{\mathrm{th}} := \sqrt{\frac{1+4c(1-2c)}{(1-2c)^2}} \cdot \sigma_\delta \cdot \max\left\{\frac{2\widetilde{D}}{\theta}, \frac{1}{\sqrt{H_1}}\right\}.$$

$f^\star$-Free Self-Tuning Schedules: SGD

Theorem 4 (Guarantees of SGD under the Self-Tuning Warmup Schedule)

Suppose Assumptions 1–2 hold and fix $c \in (0, 1/36]$. *Let* $\{\mathbf{w}_k\}$ *be generated by SGD using the schedule as in Definition 3,*

$$V_T := \sum_{k<T} \eta_k^2 \leq \frac{\widetilde{D}^2}{1536\sigma_\delta^2 \ln(2/\delta)},$$

with

$$T_w \geq 1 + \left\lceil \frac{11}{6\alpha c} \cdot \left(\frac{H_0}{2\Delta_{\mathrm{th}}} + H_1 \log \frac{U}{2\Delta_{\mathrm{th}}} \right) \right\rceil = \mathcal{O}\left(\frac{1}{\alpha c} \cdot \left(\frac{H_0}{\sigma_\delta \cdot \alpha^{-1/2}} + H_1 \log \frac{U}{\sigma_\delta \alpha^{-1/2}} \right) \right),$$

where the saturating schedule achieves a logarithmic speedup, instead of linear as in Theorem 2, in the initial suboptimality gap estimate U *again. Then, there is an event* $\mathcal{E}$ *with* $\mathbb{P}(\mathcal{E}) \geq 1 - \delta$ *such that* $\Delta_{T_w} \leq 2\Delta_{\mathrm{th}}$.

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